

On Social Evolution: A Critique and Toward a New Synthesis

Yang Zaoru

Abstract: This article inherits and critiques the Social Evolutionary Paradigm (SEP) proposed in *On Social Evolution: Phenomenon and Paradigm*. SEP treats ideas as genotypes and human social behavior, as well as social outcomes such as technologies, institutions, cultures, and political entities, as phenotypes; it constructs a mechanism of variation, selection, and inheritance; and it integrates eleven foundational paradigms, thereby providing the social sciences with a systematic evolutionary perspective. However, because SEP treats ideas as genotypes independent of DNA, it separates social evolution from the factual basis of biological evolution, leading to ambiguity in the concept of "species," theoretical redundancy, and the risk of exogenous explanation. In response, this article proposes a "Social Evolutionary Synthesis" (SES), which redefines ideas and behavior as phenotypes of agents, regards social outcomes such as technologies, institutions, cultures, and political entities as agents' extended phenotypes, and treats social phenomenon such as war as part of an environment external to agents that has a selective character. By using phenotypic plasticity to explain the rapid transformation of ideas and by unifying artificial and natural environments, SES preserves the core mechanisms of SEP while achieving a genuine ontological and mechanistic unification of social evolution and biological evolution. It thus offers a more parsimonious and coherent theoretical perspective for understanding social evolution.

Keywords: Social Evolutionary Paradigm; Social Evolutionary Synthesis; biological evolution; social evolution

I. A Review of *On Social Evolution*

On Social Evolution (hereafter OSE) is an academic work written by Professor Tang Shiping of Fudan University and published by Routledge in 2020. OSE is Tang Shiping's systematic attempt to explain human society by means of the Social Evolutionary Paradigm (hereafter SEP) and its mechanisms. The book has been regarded as the most complete and advanced attempt to date in the field of international political science to employ evolutionary thinking.^① In the endorsements preceding the preface, several scholars with major influence in international political science, such as Jervis and Katzenstein, speak highly of the book, describing it as "inspirational," "a profound study and powerful argument," and a work that "all our colleagues should read carefully." They also hold that SEP "rightly states the theory of evolution" and can be applied to micro-, meso-, and macro-level analysis in the social sciences.

^①Garlick J. *Evolution in International Relations*. Cambridge University Press; 2025.p.38.

The high praise that OSE and its attempt to propose a paradigm or theory have received from some colleagues in international political science may have two principal reasons. First, it was among the earlier works to use a scientific theory in a systematic way to unify major phenomena in human society, such as conflict and cooperation at the individual and state levels (war and peace) and development and stagnation/decline, through a relatively sophisticated or complex set of mechanisms. By minimizing exogenous explanation as far as possible, this set of mechanisms is intended to explain major phenomena in human society or international political science essentially endogenously through SEP (at least as the author states).^① This attempt to propose concrete mechanisms may allow major phenomena in human society to be broadly unified on the basis of a foundational paradigm, thereby helping readers understand human society at the level of its underlying logic. Second, on this basis, OSE adopts the perspective of SEP to review the social sciences that, from Lamarck to the present, are regarded within the SEP perspective as related to evolutionary theory or social evolution; it evaluates and criticizes them and, from a commanding vantage point, reviews major social-scientific theories up to the present, thereby establishing a dialogue with them.^② This review also helps readers, whether or not they are already familiar with contemporary social-scientific theory, to become acquainted with these theories relatively quickly and to understand their contributions and limitations in depth.

More specifically, OSE is divided into seven chapters. In Chapter 1, OSE first argues that evolutionary theory since Darwin is of extreme importance for understanding the biological world.^③ Through the mechanism of variation, environmental selection, and inheritance, Darwin unified the phenomena of the biological world in *On the Origin of Species*. As part of the biological world, human beings are the object of the science that studies human society, namely the social sciences. The social sciences must therefore take this legacy seriously; otherwise, they will fail to develop a comprehensive and fundamental understanding of human society and will remain superficial or incomplete.^④ Immediately thereafter, in Chapter 2, OSE reviews the history of biological evolutionary theory beginning with Lamarck:^⑤ from Lamarckian inheritance of acquired characteristics, to Darwin's mechanism of "variation-selection-inheritance," to neo-Darwinism/the Modern Synthesis, and then to the Extended Synthesis. OSE points out that, unlike Darwin, Lamarck misunderstood biological evolution: he held that variation in biological traits occurs after environmental selection, namely that organisms

^①Tang Shiping, *On Social Evolution:Phenomenon and Paradigm*,Routledge,2020.pp.5-8,pp.192-197.

^②Tang Shiping, *On Social Evolution:Phenomenon and Paradigm*,Routledge,2020.pp.117-170.

^③Although Lamarck was a precursor of evolutionary theory, he misunderstood its mechanism; see below.

^④At the beginning of the original OSE, Tang alludes to the words of Nobel laureate Hermann Muller: "one hundred years without Darwinism are enough." See Tang Shiping, *On Social Evolution:Phenomenon and Paradigm*,Routledge,2020.vi. OSE likewise holds that SEP can fundamentally explain all important phenomena of human society; see Tang Shiping, *On Social Evolution:Phenomenon and Paradigm*,Routledge,2020.p.25.

^⑤OSE holds that vague evolutionary ideas appeared in even earlier ancient periods, for example among ancient Chinese philosophers such as Laozi and ancient Greek philosophers such as Heraclitus, but that these were not evolutionary theory in the true sense. See Tang Shiping, *On Social Evolution:Phenomenon and Paradigm*,Routledge,2020.p.22.

actively choose trait variation under environmental pressure or inducement and then transmit it directly to their offspring. This is the so-called principle of "use and disuse" or "inheritance of acquired characteristics."^① Modern biology, however, has shown that this understanding is mistaken. Darwin correctly proposed that variation in biological traits is primarily random rather than active, that it occurs before environmental selection, and that organisms cannot subjectively choose which traits they possess in order to pass selection. Only thereafter are they selected by the external environment. Traits that permit survival or reproduction are preserved under external environmental selection and can ultimately persist; through inheritance across multiple generations, new species may form. It should be noted that Darwin at the time could not yet understand in detail the specific mechanism by which traits filtered by the environment are transmitted to offspring. Only through Mendel's research on inheritance, Weismann's discovery of the central principle (the Weismann barrier), and the discovery of the DNA double helix did Darwin's mechanism of inheritance come to be understood as the transmission of parental information by genes; hence it came to be called "neo-Darwinism." Gene-centered neo-Darwinism/the Modern Synthesis gradually became established: genetic variation leads to changes in traits (phenotypes), environmental selection follows, and genes are inherited. For a time, this became the dominant understanding or paradigm in the biological sciences. Yet the Modern Synthesis has also begun to change, mainly under the impact of epigenetics and niche-construction theory. More specifically, new findings in the biological sciences show that changes in offspring traits do not necessarily derive from genetic mutation; there are at least four pathways. First, the environment can affect parental DNA methylation and histone acetylation, thereby altering gene expression and influencing developmental traits in offspring. Second, prions can alter traits by affecting human protein synthesis, and such trait changes may be inherited by descendants. Third, parental effects mean that when certain somatic traits of parents are transmitted to offspring through reproductive materials, they may become part of the environment in which offspring genes develop, affecting gene expression and thereby changing offspring traits. Finally, in niche construction, agents (usually parents) transform the environment; the transformed environment is then left behind and affects the selection pressures faced by offspring, thereby changing traits. This is "ecological inheritance." These four new findings in the biological sciences have led the Modern Synthesis, once dominant in the field, to move on its original foundation toward a new Extended Synthesis/Extended Evolutionary Synthesis, abandoning a thoroughgoing gene-centrism.^② The Extended Evolutionary Synthesis has already exerted a certain influence in the biological sciences.^③

^①A classic example is Lamarck's view that giraffes' necks became longer because giraffes chose to stretch their necks in order to reach leaves. This acquired trait was then directly transmitted to offspring and, through inheritance across multiple generations, produced the present giraffe population.

^②Not all biologists support this view. Many biologists hold that the Modern Synthesis can explain the new findings above and need not be changed. Although many biologists also support the Extended Evolutionary Synthesis and hold that the Modern Synthesis requires revision, Nature once invited biologists on both sides of this issue to state their positions. See Laland, K., Uller, T., Feldman, M. et al. Does evolutionary theory need a rethink?. *Nature* 514, 161-164

On this basis, OSE introduces the concept of social evolution and emphasizes that, because evolutionary theory is not immutable, our understanding of evolution cannot remain confined to the perspective of gene-centrism or the Modern Synthesis. That is, the mechanism of variation, selection, and inheritance is not limited to a gene-centered perspective. Understanding human society must therefore maintain a certain distance from gene-centrism.

Chapters 3, 4, and 5 constitute the core of OSE. Chapters 3 and 4 systematically expound the mechanisms of SEP, its explanation of human social phenomena, and its relationship to other social-scientific theories and philosophies of social science. Chapter 5 then uses SEP as an analytical perspective to criticize other social-scientific theories that are regarded as having an evolutionary character.

Following Chapter 2, Chapter 3 argues that human beings are entirely different from other organisms. Although humans themselves are part of the biological world and therefore are not wholly detached from their own biological physiological/physical foundations, biological evolution, that is, the variation, environment, and inheritance of DNA, still affects the evolution of human society. Yet because human beings possess "ideational forces" not found in other organisms and can understand and create complex ideas, such as language, knowledge, culture, customs, and even institutions, technology, and culture, a purely biological-evolutionary perspective on human beings is necessarily limited. It is therefore necessary to understand them through "social evolution," that is, to understand human society by treating human ideas as genotypes and applying a mechanism of variation, environmental selection, and inheritance. Social evolution and biological evolution are two relatively separate evolutionary mechanisms. In SEP, the genotype of social evolution is no longer DNA but human ideas. Through newly generated ideas (genetic variation), human beings construct culture, institutions, and states based on those ideas as phenotypes; these are filtered by the environment and are then inherited. It should be noted, however, that in SEP the environment differs from the environment in biological evolution. It refers not only to the natural environment but, more importantly, to the artificial environment: human-made phenomena such as war and power constitute the environment of social evolution. Ideas that pass selection, along with phenotypes such as culture, institutions, and states constructed on their basis, obtain survival and inheritance.^① It should also be noted that, although they face environmental selection, ideas differ from DNA. Ideas are relatively indeterminate and can be transmitted and inherited by descendants without passing through genetic material. Social evolution is therefore Darwinian evolution nested within super-Lamarckian evolution. Here, the mechanism of the social-evolutionary paradigm is not purely Darwinian: it does not follow only (ideational) variation-(artificial) selection-inheritance, but may also include (artificial) selection-(ideational) variation-inheritance. Moreover, social evolution

(2014).

^①SEP holds that, unlike biological evolution, in social evolution genotypes can also be directly selected by the environment. See Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp. 50-53.

occurs at four clearly defined levels. The first is the individual level, namely the coevolution of biology and ideas/behavior. The second is the level of groups and organizations, where group identity and intergroup competition (war) profoundly shape the evolution of political and economic organizations, from tribes to states. The third is the level of institutions/cultural traits/knowledge, namely the process by which ideas prevail in power struggles and become solidified as institutions, morality, religion, and scientific laws. The final level is the social-system level: macro transformations from regional to global systems, such as the evolution from the premodern international system to the Westphalian system and the establishment of the capitalist system.

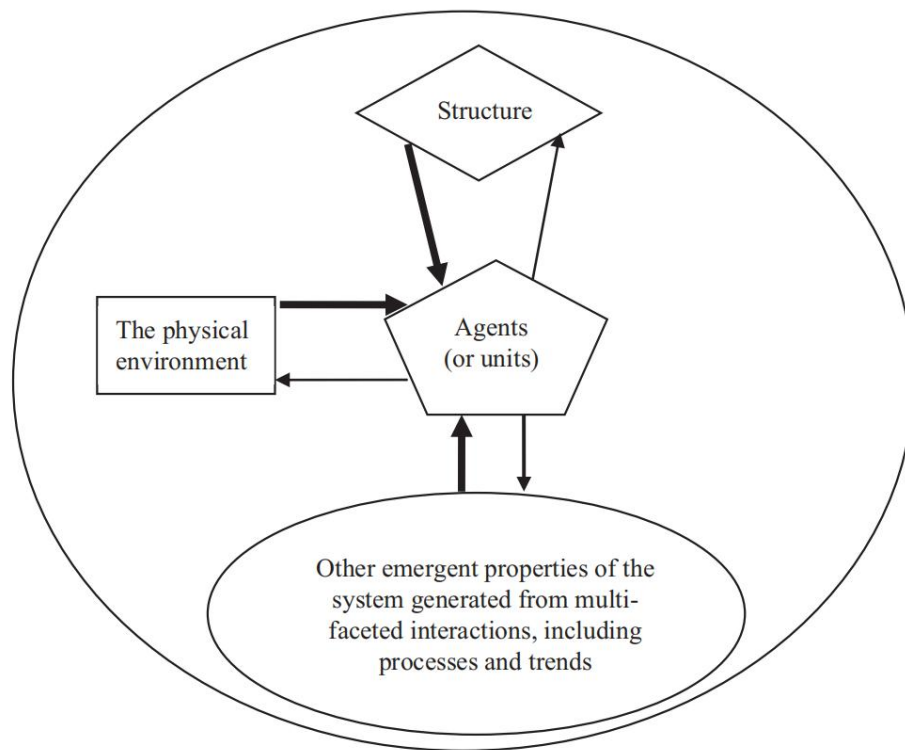


Figure 1: The Core Mechanism of the Social Evolutionary Paradigm (SEP)^①

At the end of Chapter 3, SEP argues that, in social evolution, there are at least six types of interaction between agents and the system/environment. First, there is physical constraining and enabling: geography and climate determine the distribution of agrarian civilizations, and the ocean both obstructs transportation and, after the emergence of navigation technology, facilitates globalization. This constitutes the most basic foundation. Second, there is learning, including "negative learning," by which agents draw lessons from their own failures, and "positive learning," by which they imitate

^①Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, p11.

success (what constructivism calls socialization). Collective learning is, in essence, an evolutionary process. Third, there is both material and ideational constraining and enabling: social institutions, norms, allies, and enemies both constrain certain actions of agents and enable them to take particular actions by using rules. Fourth, there is artificial selection: in its strong form, agents (states, firms) are eliminated and destroyed by the system, as in conquest or bankruptcy; in its weak form, agents are punished for violating rules or misjudging the environment. Fifth, there is constituting/constructing: agents actively internalize social ideas, identities, and norms, or they are unconsciously permeated and shaped by systemic discourse/power, in the Foucauldian sense of discipline. Finally, there is anti-socialization, or ideational resistance. This is a crucial dimension neglected by classical functionalism. Wherever power and discipline exist, agents may become aware of oppression and engage in conscious resistance and struggle. Anti-socialization is the dialectical product that socialization necessarily generates, and it is an important force driving systemic change.

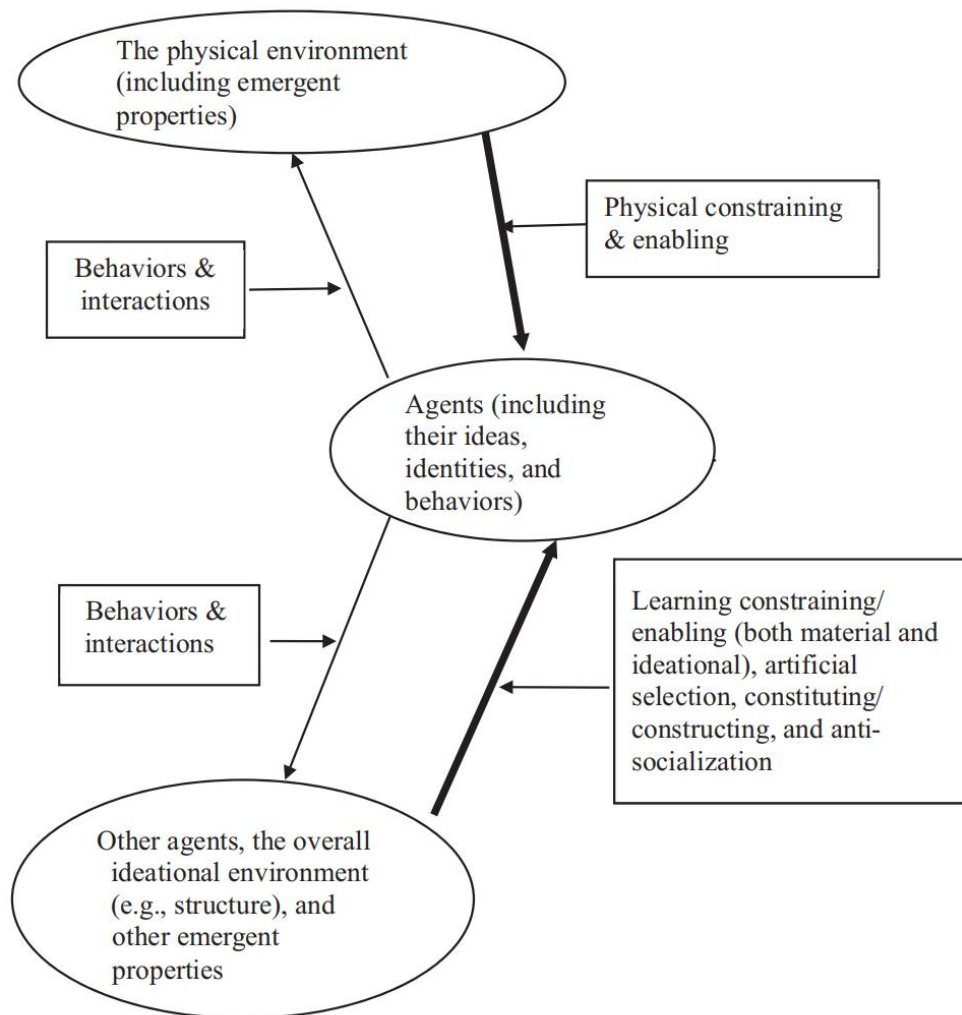


Figure 2: The Six Relationships between Agents and the System in SEP^①

In Chapter 4, SEP begins a dialogue with other social-scientific theories, as well as with social-scientific and philosophical thought. OSE seeks in this chapter to establish the status of SEP as the ultimate paradigm.^② SEP holds that the social sciences contain only eleven foundational paradigms. Earlier social science at most included ten of these paradigms, or only some of them: materialism and ideationalism, individualism and collectivism, biological determinism, socialization, and anti-socialization, and the conflict paradigm and harmony paradigm. Although the Social System Paradigm (SSP) proposed by Jervis had already incorporated the preceding nine paradigms before SEP, from the standpoint of SEP, SSP omitted the temporal dimension; SEP remedies this omission by emphasizing evolution. By incorporating eleven paradigms, SEP accomplishes what previous social science could not, or, in other words, passes the "eleven trials" and becomes the fundamental paradigm of contemporary social science.^③ In Chapter 4, OSE also clarifies possible misunderstandings of SEP. First, SEP does not mechanically apply biological evolution to society; social evolution is Darwinism nested within super-Lamarckianism, and artificial selection is far more important than natural selection. Second, it is not biological evolutionary determinism or reductionism; it rejects the attempt by sociobiology and evolutionary psychology to reduce human behavior entirely to genes. Third, it is neither a metaphor nor a simple analogy. SEP is a "generalization" of evolutionary theory to complex group systems, but it is not "generalized Darwinism." Fourth, organicism is not SEP; treating society as a (super-)organism, as in Spencer and Parsons, is the hallmark of pseudo-evolutionary theory. Human social systems consist of complex interactions between agents and the environment/system. Fifth, SEP is not social Darwinism or "survival of the fittest." In the biological world there is only survival of the relatively fitter, not of an absolute "fittest." Social Darwinism is not only biologically wrong but also, morally, a root cause of geopolitical expansion and the Nazi Holocaust. Sixth, SEP is not merely a theory of human nature. Human nature is only the background of social evolution, and obsession with it leads to reductionism. Seventh, SEP is not the "unfolding of destiny" or "historical necessity"; evolution contains contingency and rejects historical determinism. Eighth, SEP is not merely a matter of identifying and classifying stages. Simply dividing stages, such as tribe-state, is descriptive only; to count as social-evolutionary theory, one must explain how stages are transformed through the mechanisms of artificial variation-selection-inheritance or selection-variation-inheritance. Ninth, SEP is not naive adaptationism or functionalism. One cannot assume that every institution existing in society is "good" or "perfectly adapted," because many poor institutions are maintained by power. Tenth, SEP does not deny human intentionality and agency; on the contrary, human intentionality is central to artificial selection. Eleventh,

^①Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, p.108.

^②Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp.95-96.

^③See Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp.96-101.

SEP is not cultural determinism. Culture itself is a product of social evolution and cannot determine itself in isolation from material foundations. Finally, an evolutionary theory lacking an analysis of "power" is inadequate. Power is the core selection mechanism of social evolution, and theories that ignore power, such as some evolutionary economics and evolutionary game theory, are bound to remain incomplete.^①

After Chapter 4's exposition and clarification of SEP as a paradigm, OSE uses SEP as an analytical perspective in Chapter 5 to criticize all social-scientific theories that SEP regards as possessing an evolutionary coloration. For example, Spencer's social Darwinism equated evolution with teleological progress and the unfolding of design. Spencer confused biological "development" (the unfolding of design in an individual embryo) with "evolution" (variation in populations through selection). This led him to the mistaken belief that social evolution is a necessary, unilinear progression toward perfection. At the same time, Spencer regarded society as an organism and understood social dynamics as "differentiation and integration" carried out to meet the functional needs of that organism. This later developed into the "structural functionalism" that dominated sociology and anthropology (Durkheim, Parsons, Merton). Functionalism holds that "what exists is rational/optimal" and uses "harmony" to cover up the harsh "selection and conflict" within evolution; it is therefore pseudo-evolutionary. In addition, social Darwinism takes "survival of the fittest" as a central principle. Spencer coined the phrase, and Darwin unfortunately adopted it. Logically, this points toward "survival of the very fittest," leading to racism, eugenics, and Nazi geopolitics.^② In addition to social Darwinism, sociobiology and evolutionary psychology are likewise criticized by SEP. These two fields are regarded as "reductionist," with a tendency to reduce the evolution of human society to gene-centrism or natural selection.^③ Other social scientists, such as Hayek and the Austrian School, opposed rational design and argued that existing rules are the optimal results of spontaneous evolution. SEP criticizes this as a form of "naive adaptationism." Hayek refused to acknowledge the role of "intentional design" in institutional change and entirely ignored the decisive place of "social power" and conflicts of interest in shaping order. Hayek's theory ultimately degenerated into a conservatism that defended the status quo, rather than a scientific evolutionary theory. Likewise, although some social scientists display an evolutionary coloration, they emphasize a "historical necessity" and a linear view of production and social development, which remains Spencerian teleology rather than Darwinian blind variation and selection.^④ SEP also criticizes the generalized Darwinism proposed by Hodgson and Knudsen in *Darwin's Conjecture*, published in 2010. Although it regards this as a useful attempt, SEP points out that, with the discoveries associated with biology's "Extended

^①Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp109-114.

^②Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp122-127.

^③Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp127-132.

^④See Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp134-138.

Synthesis" (epigenetics, among others), even biological evolution is no longer purely Darwinian. In the domain of social ideas, evolution is even more typically "super-Lamarckian." Generalized Darwinism attempts to downgrade human "artificial selection" and subordinate it to "natural selection"; this is a mistaken reductionism. More regrettably, as economists, the proponents of generalized Darwinism still fail to incorporate "social power" into the core mechanism of social evolution. What we need, therefore, is not generalized Darwinism but generalized evolutionism, and SEP is its standard form.^① In Chapters 6 and 7, OSE reiterates the importance of SEP, arguing that OSE can explain cultural change, institutional change, the Industrial Revolution, the institutional foundations of economic development, and the evolution of international politics within the international system. It calls on social scientists to study social-scientific questions from a more evolutionary perspective.^②

In sum, by expounding a relatively complex set of mechanisms and a paradigm in the form of SEP, OSE plays an important role in enabling readers and fellow social scientists to understand social phenomena and the social sciences more deeply. SEP's summary and synthesis of human society and of social-scientific theories up to the present, together with its concrete attempt to propose mechanisms that explain the phenomena above endogenously, constitute the most advanced systematic attempt currently available. The praise that international political scientists such as Jervis and Katzenstein have given OSE is by no means groundless. Nevertheless, SEP has limitations, the most important of which is that it divides the relationship between biological evolution and social evolution.

II. A Critique of OSE: From the Social Evolutionary Paradigm (SEP) to the Social Evolutionary Synthesis (SES): Toward a New Synthesis

As noted above, although SEP makes important contributions to understanding human society and the social sciences, it still has limitations. One relatively evident limitation is its possible anthropocentric tendency. SEP assumes that only human beings possess ideas; however, other nonhuman organisms may have psychological activities that humans cannot understand, or may communicate in ways that humans cannot understand. Moreover, human beings themselves have long lived in and transformed the natural world and are an inseparable part of it. Finally, although SEP emphasizes the constraints imposed on humans by the physical environment, it consistently stresses that the artificial environment is the principal environment of social evolution. It is therefore difficult to understand the effects of nonhuman actors, such as other species, viruses, new technologies, and nature, on human society and on human beings.^③ In addition, SEP claims that works in sociobiology and evolutionary psychology are reductionist because

^①Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp. 160-162.

^②See Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp. 171-190, pp. 201-204.

^③Xiao Alvin Yang, *On social evolution: phenomenon and paradigm*, *International Affairs*, Vol 97(5), September 2021: 1622-1624.

they focus entirely on the biological dimension of human behavior and therefore do not integrate the influence of society on human behavior. Sociobiologists such as Alcock have advanced a different view: they argue that sociobiology and the propositions of the social sciences are not incompatible. More precisely, sociobiology and the social sciences do not address the same level of explanation. According to this view, it is necessary to distinguish between proximate and ultimate levels: the former concerns environmental or cultural influences, while the latter concerns the evolutionary dimension of behavior. SEP appears to understate the influence of biological evolution on social evolution.^①

More importantly, SEP's principal limitation lies in the substantial distinction it draws between social evolution and biological evolution, which results in a factual separation of the two. Because SEP emphasizes ideas as the genotype of social evolution, even though it repeatedly declares that social evolution must rest on biological evolution as its material foundation, it becomes difficult to describe or explain the mechanism of interaction among human ideas, the artificial environment, and human DNA. As shown in Figures 1 and 2, the effects of the artificial environment and the natural environment on agents/units (human beings) are distinguished from one another and are essentially disconnected. SEP also provides no account of any mechanism through which the artificial environment acts on human genes. These problems force SEP, when expounding mechanisms, to divide ideas and DNA into two evolutionary mechanisms. As a result, social evolution is to a very large extent difficult to connect with biological evolution, which produces theoretical redundancy in SEP itself and reveals a limitation in parsimony. At the same time, and perhaps more crucially, because it treats ideas as genotypes independent of human genes, SEP is compelled to regard biological evolution and social evolution as two distinct mechanisms. Although SEP claims that it is by no means a metaphor and is instead a correct application of evolutionary theory, and although evolutionary theory is a fundamental and correct explanation of the laws of biological evolution, SEP lacks a mechanism connecting biological evolution and social evolution. Its use of evolutionary theory is therefore at least incomplete (if not mistaken), and the mechanisms and relationships between social evolution and biological evolution can only become metaphorical or conjectural. Moreover, because SEP insists that ideas are the genotype of social evolution, it not only creates difficulties of understanding for other scientists,^② but also requires social evolution to reject evolutionary psychology and a large body of empirical findings that apply evolutionary theory to human problems. This creates substantial obstacles to the expansion and extension of SEP.^③ In addition, the emergence of new ideas is understood as random variation analogous to DNA mutation. This makes ideas in SEP highly vulnerable to becoming an exogenous explanation, contradicting SEP's own aim of explaining human society as endogenously

^① Alexandre Tawil: "Shiping Tang, On Social Evolution: Phenomenon and Paradigm". *OEconomia*, Vol. 12 (3): 731-737. [Open Economica].2022

^② Almost all biologists regard DNA as at least the genotype of human beings in evolutionary theory, whether in biological evolution or in social evolution.

^③ Garlick J. *Evolution in International Relations*. Cambridge University Press; 2025.p.38.

as possible. Furthermore, because SEP treats social evolution as an evolutionary mechanism relatively independent of biological evolution, it has to confront new problems. For example, social evolution must solve the problem of speed: in biological evolution, evolutionary change is slow, whereas in social evolution it may be extremely rapid, because the emergence of new ideas and changes in the social environment are extremely fast relative to the pace of biological evolution. SEP can therefore only assert or assume that ideas, as genotypes, naturally possess very high speed; further explanation is lacking.^① In order to account for the fact that ideas can be transmitted directly to descendants and to others without passing through genetic material, SEP must also posit that social evolution is "Darwinian evolution nested within super-Lamarckian evolution," which further increases the redundancy of SEP as a theory (if not its error). In addition, because SEP assumes that social evolution is independent relative to biological evolution, it must also answer what counts as a "species" in social evolution. SEP is therefore compelled to assume or hold that occupations, groups, cultural systems, and knowledge systems are the "species" of social evolution. In biological evolution, the formation of a new species has a clear definition.^② In social evolution, however, the definition of "species" is relatively vague. More importantly, the boundaries among species are even more ambiguous, because no standard exists. From the SEP perspective, the "species" of social evolution cannot possibly involve reproductive isolation. SEP therefore has to acknowledge that, in social evolution, the definition or demarcation of its so-called "species" can only be vague.^③ This makes social evolution in the SEP perspective increasingly remote from evolutionary theory as generally understood by biological scientists (if not mistaken), and the vagueness of its definitions also risks criticism for lacking scientificity. Moreover, this in fact exposes SEP's claim to uphold the priority of materialism to possible doubt.^④

Therefore, in order to resolve the limitations of SEP described above, it is necessary to make a modest adjustment to SEP. A feasible approach is to keep human beings, whether individuals or groups, unchanged as agents, and to unify the artificial environment and the natural environment under the single concept of environment. Ideas should be adjusted from genotypes to phenotypes; organs and behavior should likewise be treated as phenotypes; and technologies, cultures, institutions, and political entities in human society, such as communities, tribes, city-states, and states, should be called human extended phenotypes.^⑤

^①Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp.72-73.

^②This means that after biological genetic mutations are inherited across multiple generations, reproductive isolation from the original species gradually forms; this is the birth of a new species. See 《新物种诞生的必经之路：生殖隔离》，浙江大学生物演化中心，2022年12月2日，<https://evolution.zju.edu.cn/science/detail21.html>.

^③See Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp.66-68.

^④See Tang Shiping, *On Social Evolution: Phenomenon and Paradigm*, Routledge, 2020, pp.96-97.

^⑤The extended phenotype is relative to the phenotype. The phenotype refers to the traits formed when genes develop into an organism. The extended phenotype refers to the "traits" of an organism that are formed through that organism's transformation of what lies outside the organism itself; that is, its extended phenotype. See Richard Dawkins: *The Extended Phenotype: The Gene as the Unit of Selection*, Oxford University Press, 1981, pp.179-210.

Here, two fundamental concepts in social evolution are established: environment and agents/units. In social evolution, environment, agents, and the outcomes produced by their interactions are therefore the three constitutive elements; the relationships among them constitute the entirety of social evolution. On the basis of preserving all of SEP's claims about the complex mechanisms, ontology, methodology, and epistemology of social evolution, this understanding will greatly reduce the redundancy in the SEP perspective's account of social evolution. This new understanding may also genuinely connect the mechanisms of social evolution and biological evolution, unify social evolution with biological evolution, and form a new paradigm or synthesis for understanding human society and the social sciences: the Social Evolutionary Synthesis (SES).

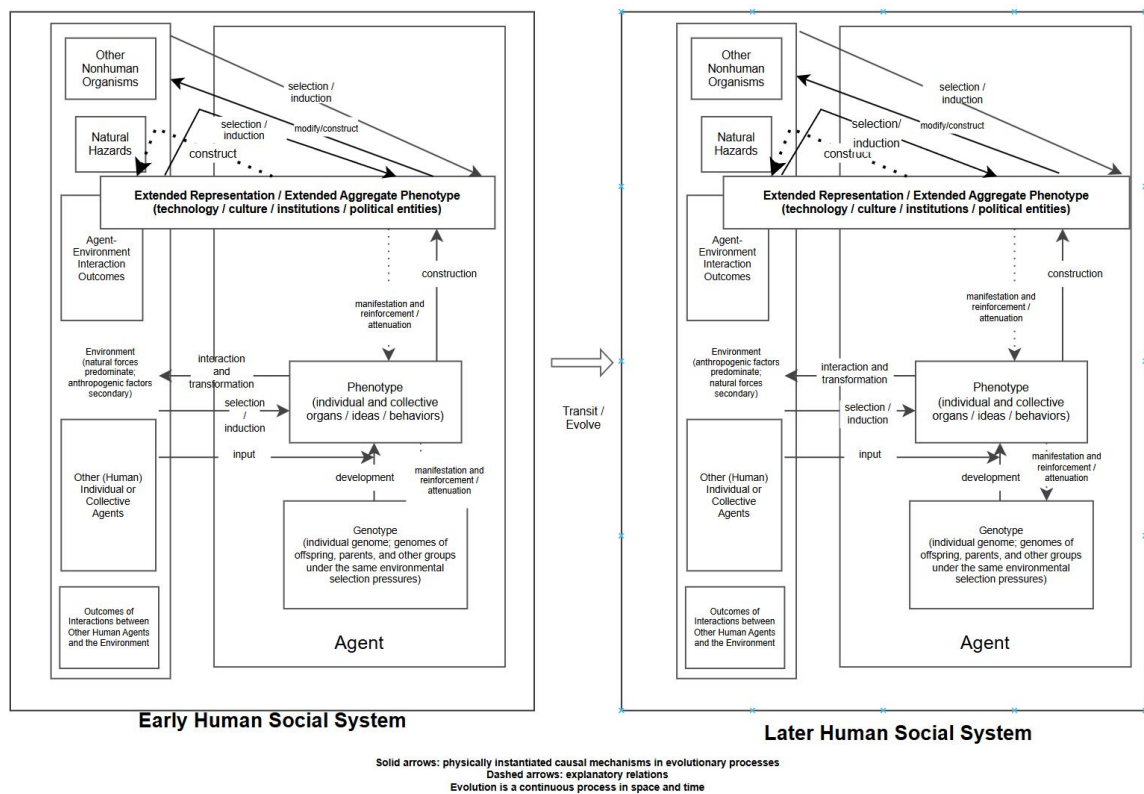


Figure 3: Mechanism of the Social Evolutionary Synthesis (SES)

As Figure 3 shows, distinguishing the environment into an artificial environment and a natural environment may be analytically convenient, but ontologically the two should not differ, because both are, in essence, environments external to human agents that exert selection pressures on them. Since the environment itself comprises both artificial and natural environments, other nonhuman organisms, natural disasters, other human individual or collective agents, and the results of interactions between agents themselves and the environment, such as war, are all possible environments that may impose selection pressure on agents. At the same time, in SES, agents are the same as in SEP: human beings as individuals or collectives, because all may face selection pressures

from the external environment. In SES, however, the genotype is consistent with biological evolution: it is the gene pool of individuals and of groups, including offspring, parents, and others who face the same environmental selection pressure.^① Under environmental inputs or influences, such as nutrient intake, radiation, or physical injury, genotypes develop into phenotypes, such as organs. In any healthy human being, the organ of the brain necessarily produces certain psychological-activity traits, ideas, and behaviors. In SES, ideas are phenotypes rather than genotypes.^② Here, as in SEP, SES acknowledges that human beings differ significantly from other organisms because of ideas;^③ social evolution, therefore, has its own distinctive mechanisms. Yet this distinctiveness does not mean that social evolution is not part of biological evolution. Because agents' behavior, like ideas, is likewise an observable characteristic or trait of organisms and exists within the organism, it is also a human phenotype. The distinctive phenomena in human society that SEP regards as the "phenotypes" of ideas, namely technologies, cultures, institutions, political entities, and the like, are in SES extended phenotypes, because they exist outside the agent's organism and manifest traits of the agent. It should be noted that in SES extended phenotypes are like phenotypes: although they often help agents adapt, they are not always adaptations. They may also be maladaptive phenotypes that reduce agents' fitness.^④ Through transforming the environment, agents form extended phenotypes and engage in niche construction. It should be emphasized that extended phenotypes and niche construction are not identical; the former is a subset of the latter. Both extended phenotypes and niche construction are results of agents' transformation of the environment, and both may exert selection pressure on agents. For a particular niche construction to count as an extended phenotype, it must satisfy at least three conditions: it must be an observable trait with some degree of stability; it must express the agent's phenotype/traits; and it must fall within the agent's controllable scope. For example, although human-made plastic waste creates selection pressure as part of the environment, it does not express human phenotypes/traits, and because it is globally distributed, it lacks a controllable scope for agents; it is therefore

^①Multilevel selection theory holds that both individuals and groups may be filtered by the environment, not only kin selection. Multilevel selection theory has become accepted in the biological sciences and is supported by a substantial body of empirical evidence. See Damuth, J., Heisler, I.L. Alternative formulations of multilevel selection. *Biol Philos* 3, 407-430 (1988). <https://doi.org/10.1007/BF00647962>.Marín C, Clark AB, Philson CS, Eldakar OT and Wade MJ (2026) Abundant empirical evidence of multilevel selection revealed by a bibliometric review. *Front. Ecol. Evol.* 14:1752597. doi: 10.3389/fevo.2026.1752597

^②A phenotype refers to observable characteristics of an organism, such as morphological or biochemical characteristics and the presence of a disease or condition. A phenotype is jointly determined by the expression of an individual genotype and the influence of environmental factors, such as nutritional status and climate. Ideas, in substance as human psychological activities, fit this definition. See "APA Dictionary of Psychology", American psychology association, <https://dictionary.apa.org/phenotype>. Phenotype: Psychology Definition, History & Examples, Zimbardo, <https://www.zimbardo.com/phenotype-psychology-definition-history-examples/>.

^③This refers only to the fact that human beings generate psychological activities distinctive from those of other organisms. SES does not imply that other organisms cannot generate complex psychological activities.

^④Many empirical studies show that extended phenotypes may likewise be unfavorable to biological fitness. See Jordan LA, Maguire SM, et al. (2016). The social and ecological costs of an 'over-extended' phenotype. *Proceedings of the Royal Society B: Biological Sciences*, 283(1822), 20152359. Ecological and evolutionary traps Schlaepfer, Martin A. et al. *Trends in Ecology & Evolution*, Volume 17, Issue 10, 474 - 480

not a human extended phenotype. By contrast, a state, as a form of niche construction through which agents transform the environment and affect the selection pressures they face, is an observable trait with some degree of stability; it also expresses the phenotypes/traits of the state's citizens, such as ideas, thoughts, and behavior; and it has a scope.^① In addition to these three conditions, extended phenotypes possess a duality: they are both part of the environment and part of the agent. Accordingly, they are also a type of phenotype. Through selecting agents' extended phenotypes, an environment external to agents affects the fitness of their phenotypes and genotypes. For example, after a state is selected by war external to itself and is eliminated (destroyed),^② its citizens may also be imprisoned or massacred. And niche construction does not always hold this trait, for niche construction can not necessarily be extended phenotype by definition.

However, extended phenotypes differ slightly from phenotypes. Because extended phenotypes are traits outside the agent's organism, the environmental selection and elimination of an agent's extended phenotype does not necessarily mean that the agent cannot survive and reproduce. For example, after Sikkim was annexed by India, although some Sikkimese citizens still died in the process, the survival of all Sikkimese citizens was not threatened on a large scale.^③ In addition, agents' extended phenotypes may also transform the environment and thereby form new extended phenotypes or extended composite phenotypes, changing the agents' capacity to adapt to the environment.^④ For example, in the early period of the founding of the United States, the different states, by cooperating and by transforming the environment external to each state, formed a larger federation: a new and larger extended phenotype or extended composite phenotype. At the same time, because extended phenotypes/extended composite phenotypes are also part of the environment, they likewise affect agents' phenotypes or extended phenotypes. In SES, changes in ideas (i.e., genotypes), institutional change, cultural change, economic development and stagnation or decline, and changes in the international system/systemic order, all of which SEP regards as social evolution, are changes in phenotypes, extended phenotypes, and the environment; they do not involve changes in genotypes. Since human society emerged later than the human species, there is no birth of a "new species" in social evolution. Moreover, because phenotypes are highly plastic, they can change at any time in response to environmental selection pressures, thereby providing agents with

^①For related discussions, see Dawkins, R. Extended Phenotype - But Not Too Extended. A Reply to Laland, Turner and Jablonka. *Biology & Philosophy* 19, 377-396 (2004). Rouse, Joseph. *Social Practices as Biological Niche Construction*. University of Chicago Press, 2023. Chicago Scholarship Online, 2024. Tang, S. Niche construction to social sciences: can it help?. *Theor Soc* 54, 209-241 (2025).

^②In SES, the understanding of elimination is consistent with evolutionary theory and SEP. Elimination does not imply any value judgment or any directionality; it merely indicates that a phenotype/extended phenotype cannot survive or reproduce because it is not adapted to the environmental selection pressures it currently faces.

^③The extinction of Sikkim has its particularity. Before being annexed by India in 1975, Sikkim was already an Indian protectorate and had not obtained independent membership in the United Nations system. In this sense, India had already to some extent become an extended phenotype of Sikkimese citizens before 1975.

^④See Phillips, J. D. (2016) Landforms as extended composite phenotypes. *Earth Surf. Process. Landforms*, 41: 16-26. Lewis Revelly, Paul Eggleton, Rebecca Clement, Chuanyu Zhou, Tom R. Bishop; The diversity of social complexity in termites. *Proc Biol Sci* 1 June 2024; 291 (2024): 20232791.e Araújo GF, Moioi RC and de Souza SJ (2021) The Shared Use of Extended Phenotypes Increases the Fitness of Simulated Populations. *Front. Genet.*

the possibility of adapting under such pressures.^① Ideas, as human phenotypes, therefore possess a high degree of plasticity. They may at any time be learned from, or accepted from, an environment external to themselves, such as other agents including human beings, or environments transformed and left behind by other agents. The emergence of new ideas is itself a manifestation of phenotypic plasticity; there is no need to invent "Darwinian evolution nested within super-Lamarckian evolution" to explain it.

After clarifying the concepts of environment and agent in SES, it is necessary to explain the specific mechanism of SES. First, the genes of agents, whether individual or collective, develop under environmental inputs or influences into phenotypes such as organs, psychological activities (that is, ideas), and behavior. Phenotypes construct extended phenotypes such as technologies, cultures, institutions, and political entities. On the one hand, as part of the environment, such extended phenotypes exert selection effects back upon agents' own phenotypes or extended phenotypes. This selection is primarily a physical process carried out through social power. For example, a state's legal system exerts inductive and selective effects on agents' phenotypes, such as ideas and behavior; citizens who do not obey the law may be imprisoned or even sentenced to death.^② At the same time, as noted above, extended phenotypes also transform the environment, thereby changing the selection pressures that the environment imposes on agents. In the international system, such transformation may occur at the state level: for example, a state may prevent further aggression by an enemy state through a war of self-defense. It may also occur at the global level: for example, great powers may reduce the likelihood of world war by creating suprasovereign organizations such as the United Nations. At the same time, such transformation of the environment may be left behind, inducing or selecting agents' phenotypes or extended phenotypes and ultimately affecting the human genotype. It should be noted that the environment's inducement of agents' phenotypes and extended phenotypes generally occurs before its selection of them, because the plasticity of phenotypes and extended phenotypes gives them the capacity to change under environmental inducement, temporally prior to environmental selection. This does not mean, however, that such plasticity necessarily allows agents to pass environmental selection. Moreover, because human phenotypes in social evolution are highly plastic, the selection of agents' phenotypes in social evolution has only a very small effect on agents' genotypes, since human phenotypes such as ideas and behavior do not correspond one-to-one to any allele of their own.^③ Human biological evolution is therefore extremely slow.^④

^① A large body of empirical evidence demonstrates the existence of phenotypic plasticity; see Sommer R. J. (2020). Phenotypic Plasticity: From Theory and Genetics to Current and Future Challenges. *Genetics*, 215(1), 1-13, for a review.

^② A small number of empirical studies show that human beings, influenced by their own genes, develop different behaviors or psychological traits; for example, parental investment theory in mate choice. Although mate choice is very important, it is only one part of social evolution that is influenced by genotype, not the whole picture. See Robert Trivers: *Sexual Selection and the Descent of Man, 1871-1971* (pp.378), Publisher: Aldine, Chicago

^④ Human beings, however, are still evolving. See Stock, Jay T. "Are humans still evolving?." *The EMBO Reports* 9.1

Furthermore, the preceding description of the SES mechanism represents only one complete cycle of that mechanism. In fact, in SES, social evolution is a continuous process in space and time. The results of one cycle of social evolution may become the environment of agents in the next and subsequent cycles, provided that those results are not transformed or selected out. For example, political institutions, legal systems, cultural systems, and technological conditions created by parents may induce and select the ideas of their offspring.^① Offspring may develop phenotypes under the influence of earlier environments and thereby transform the environment they themselves face. For example, the creation of an institution often represents, or is more conducive to, the traits and fitness of higher-ranking individuals within the collective agent.^② This is because human beings, perhaps unfortunately, evolved hierarchy as an extended phenotype early in their emergence as a species.^③ This extended phenotype has still not been selected out by the environment. The existence of hierarchy therefore induces the extended phenotypes created by collective agents in ways that, in fact, are more favorable to the traits and fitness of higher-ranking individuals within that group. Yet although this fact exists in social evolution, agents' ideas may also transform the environment or become part of the environment external to other agents. Social evolution therefore possesses a certain degree of intentionality: human beings have the possibility of transforming society according to their own ideas and intentions, although such transformation remains constrained by the mechanism of SES and cannot occur exactly as agents' ideas imagine. Further, as noted above, agents' transformations of the environment can be inherited by offspring, and offspring's transformations of the environment can have the same effect. Transformations of the environment by agents therefore have a cumulative character, as seen in technological progress and the growth of scientific knowledge. This also means that, in social evolution, the external environment faced by agents changes from early human social systems, in which artificial factors were secondary and natural factors primary, toward systems in which artificial selection predominates while natural factors remain fundamental. This makes intentionality in social evolution increasingly prominent.

III. Conclusion

Through sophisticated mechanisms and an extensive examination, OSE systematically introduces the mechanisms of evolutionary theory into the social sciences and provides readers and fellow social-scientific researchers with a valuable study from which to draw. However, because in practice it distinguishes the mechanisms of social

(2008): S51-S54. Hawks, John. "Still evolving (after all these years)." *Scientific American* 311.3 (2014): 86-91.

^①In biology this is called ecological inheritance. See Odling-Smee, John, and Kevin N. Laland. "Ecological inheritance and cultural inheritance: what are they and how do they differ?." *Biological Theory* 6.3 (2011): 220-230.

^②唐世平：《制度变迁的广义理论》，北京：北京大学出版社，第38页。

^③See Gould, Roger V. "The origins of status hierarchies: A formal theory and empirical test." *American journal of sociology* 107.5 (2002): 1143-1178. Dubreuil, Benoit. *Human evolution and the origins of hierarchies: The state of nature*. Cambridge University Press, 2010.

evolution from those of biological evolution, SEP inevitably generates new problems: incompleteness, the risk of relying on exogenous explanation, and redundancy in the application of evolutionary theory to explanations in the social sciences. SES holds that, by clarifying its core concepts and mechanisms, it can accept the mechanisms and strengths of SEP, bring SEP's contributions into play, and at the same time overcome SEP's problems, thereby realizing the unification of social evolution and biological evolution and moving toward a new synthesis.